

# LECTURE NOTES MATH254

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## 1. LECTURE 4

In this class we introduce integral domains, a special class of rings.

**1.1. Zero divisors and cancellation.** Let  $R$  be a ring. When doing calculations in  $R$  it is often useful to be able to say that if:  $ax = ay$ ,  $a \neq 0$ , then  $x = y$ . That is, it is useful if we can 'cancel the  $a$ '. However, note that this is not always possible. As an example, consider the ring  $R = M_{2 \times 2}(\mathbb{R})$ . We consider the matrices:

$$(1.1) \quad A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix},$$

$$(1.2) \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

then:

$$(1.3) \quad CA = \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}, \quad CB = \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}.$$

That is,  $A \neq B$  and  $C \neq 0$  but  $CA = CB$ . So we can not expect, in general, that just because  $ax = ay$ ,  $a \neq 0$ , we have  $x = y$  in some ring  $R$ . Note that if  $ax = ay$  then  $a(x - y) = 0$ , so if  $x \neq y$  and  $a \neq 0$  then  $0 = a(x - y)$ . So we can capture the failure of cancellation with solutions to the equation  $ab = 0$ ,  $a, b \neq 0$ , in  $R$ .

**Definition 1.1.** If  $a, b \in R \setminus 0$  satisfies  $ab = 0$  then  $a$  and  $b$  are zero divisors (or divisors of 0).

It turns out that zero divisors is the right notion to have cancellation in a ring  $R$ .

**Lemma 1.1.** Let  $R$  be a ring. The cancellation law  $ax = ay$  implies  $x = y$  holds for all  $a \neq 0$  in  $R$  if and only if there are no zero divisors in  $R$ .

*Proof.* Assume that there are no zero divisors in  $R$ . If  $a \neq 0$  and  $ax = ay$  then  $a(x - y) = 0$ , so, since  $R$  has no zero divisors, we have  $a = 0$  or  $x - y = 0$ . But  $a \neq 0$  by assumption so  $x - y = 0$  or  $x = y$ . Conversely, if the cancellation law holds then if  $ab = 0 = a0$  so either  $a = 0$ , or we can use the cancellation law to deduce  $b = 0$ . That is, if  $ab = 0$  then either  $a = 0$  or  $b = 0$ , so  $R$  has no zero divisors. ■

**Example 1.1.** Let  $n \geq 2$ . Let  $x \in \mathbb{Z}_n \setminus 0$ , it will be convenient for the remainder to think of  $x \in \{1, 2, \dots, n-1\}$ . The element  $x$  is a zero divisor if and only if  $x$  is not coprime to  $n$ . Indeed, if  $\gcd(x, n) = d > 1$  then we have:

$$(1.4) \quad x \cdot \frac{n}{d} = \frac{x}{d} \cdot n \in n\mathbb{Z}$$

so  $(n/d) \cdot x = 0$  in  $\mathbb{Z}_n$ , but both  $x$  and  $n/d$  are non-trivial elements of  $\mathbb{Z}_n$ . Conversely, if  $x$  is a zero divisor then we find  $y \in \mathbb{Z}_n \setminus 0$  such that  $xy \in n\mathbb{Z}$ . Note that  $y < n$  so  $n$  does not divide  $y$ , it follows that at least one prime in the prime decomposition of  $n$  must divide  $x$ . This then implies that  $\gcd(x, n) > 1$ .

**Example 1.2.** An important consequence of the previous example is that  $\mathbb{Z}_p$  has no zero divisors if  $p$  is prime since any element  $\{1, \dots, p-1\}$  is coprime with  $p$  in this case.

**1.2. Integral domains.** We can now define the main object of this class *integral domains*.

**Definition 1.2.** A ring  $R$  is an integral domain if:

- (i)  $R$  is a commutative ring with unity  $1 \neq 0$ ,
- (ii)  $R$  contains no zero divisors.

**Example 1.3.** As we have seen, if  $p$  is prime then  $\mathbb{Z}_p$  has no zero divisors, so  $\mathbb{Z}_p$  is an integral domain (note that  $\mathbb{Z}_p$  is commutative and has unity). If  $n$  is not prime, then  $\mathbb{Z}_n$  is not an integral domain.

**Example 1.4.** The integers  $\mathbb{Z}$  is an example of an integral domain since  $n \cdot m = 0$  does imply  $n = 0$  or  $m = 0$  within the integers.

**Example 1.5.** If  $F$  is a field and  $ab = 0$  with  $a \neq 0$  then  $a^{-1} \cdot (ab) = a^{-1}0 = 0$  or  $(a^{-1}a)b = b = 0$ . That is,  $a$  is not a zero divisor since  $b = 0$ . Since  $F$  has no zero divisors, it follows that a field  $F$  is an integral domain.

**Theorem 1.1.** Let  $R$  be a finite integral domain, then  $R$  is a field.

*Proof.* Note that the only thing we must prove is that for each  $a \in R \setminus 0$  there is some  $b \in R$  such that  $ab = 1$ , since the remaining field axioms are part of the definition of an integral domain. Let  $a \neq 0$ . We define a map  $f_a : R \rightarrow R$  by  $f_a(b) = ab$ . We claim that  $f_a$  is a bijective map. Since  $R$  is finite, it suffices to show that  $f_a$  is injective. Now, if  $f_a(b) = f_a(b')$ , then  $ab = ab'$  or  $a(b - b')$ . Since  $R$  is an integral domain and  $a \neq 0$  we conclude that  $b - b' = 0$  or  $b = b'$ . That is, if  $f_a(b) = f_a(b')$  then  $b = b'$ , so  $f_a$  is injective (which implies that  $f_a$  is bijective, since its an injective map between finite sets with the same number of elements). Since  $f_a$  is bijective we conclude that there is some  $b \in R$  such that  $f_a(b) = 1$ , or  $ab = 1$ . ■

**Corollary 1.1.1.** If  $p$  is prime then  $\mathbb{Z}_p$  is a field.

*Proof.* Note that  $\mathbb{Z}_p$  is a finite integral domain. ■

**1.3. Characteristic of a ring.** In a ring  $R$  we can ask if there is some integer  $n$  such that  $na = 0$  for all  $a \in R$ ? That is, is there an integer  $n$  such that if we sum any element  $a$  with itself  $n$  times we get 0? Clearly this is true in  $\mathbb{Z}_n$  since if  $x \in \mathbb{Z}_n$  then  $nx = 0 \in \mathbb{Z}_n$ . However, if we consider  $R = \mathbb{Z}$  and take  $1 \in \mathbb{Z}$  then  $n \cdot 1 = n \neq 0$  for all  $n > 0$ .

**Definition 1.3.** Let  $R$  be a ring. If there exists a positive integer  $n$  such that  $na = 0$  for all  $a \in R$  then we say that  $R$  has positive characteristic, and the characteristic of  $R$  is the minimal positive integer  $n$  such that  $na = 0$  for all  $a \in R$ . If  $R$  does not have positive characteristic then we say that  $R$  has characteristic 0.

**Example 1.6.** The ring  $\mathbb{Z}_n$  has characteristic  $n$ . The rings  $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ , and  $\mathbb{C}$  have characteristic 0.

**Lemma 1.2.** Let  $R$  be a ring with unity 1. If  $n \cdot 1 \neq 0$  then  $R$  has characteristic 0. If  $n$  is the minimal positive integer such that  $n \cdot 1 = 0$  then  $R$  has characteristic  $n$ .

*Proof.* If  $n \cdot 1 \neq 0$  for all  $n \geq 1$  then  $R$  has characteristic 0 by definition (since  $1 \in R$ ). Conversely, note that if  $n \cdot 1 = 0$  then:

$$(1.5) \quad na = a + a + \dots + a = (1 + 1 + \dots + 1)a = (n \cdot 1)a = 0$$

for all  $a \in R$ . The second part of the lemma is immediate from this. ■

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